

# Decoupled Monte Carlo Tree Search for Cooperative Multi-Agent Planning

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## 1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

Decoupled MCTS where each agent executes independent tree searches combined with stochastic action-selection policies, eliminating centralized joint-action tree explosion.

## 2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

Repeated matrix games, MMDP firefighting, meeting-in-grid, warehouse commissioning benchmark.

### Key Numerical Findings & Empirical Proofs:

Achieves >10% performance gain over decoupled baselines in warehouse commissioning and matches centralized planning accuracy with lower coordination overhead.

## 3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

### Identified System Failure Modes:

Performance is highly sensitive to action synchronization heuristics; coordination overhead grows when agent interaction becomes dense.

## 4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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